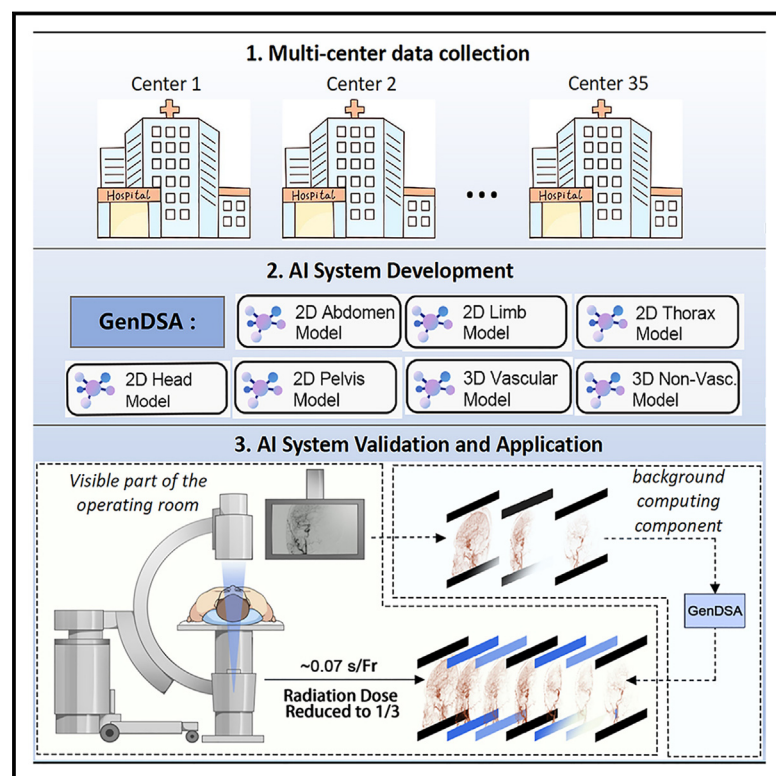


Large-scale pretrained frame generative model enables real-time low-dose DSA imaging: An AI system development and multi-center validation study

Graphical abstract



Highlights

- The proposed DV-MuRC dataset contains ~ 3 million DSA images from 27,117 patients
- The proposed GenDSA outperforms state-of-the-art video interpolation algorithms
- GenDSA reduces DSA radiation dose to 1/3 without compromising clinical requirements

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In brief

Zhao et al. developed a large-scale pretrained frame generative method (GenDSA) to realize real-time, low-dose DSA imaging. GenDSA was trained and validated by employing millions of DSA images from 35 hospitals, which demonstrated that GenDSA can achieve state-of-the-art performance and reduce the DSA radiation dose to 1/3 of clinically available protocols.

Translation to Patients

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Article

Large-scale pretrained frame generative model enables real-time low-dose DSA imaging: An AI system development and multi-center validation study

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CONTEXT AND SIGNIFICANCE Digital subtraction angiography (DSA) devices have been commonly used in hundreds of different procedures, requiring multiple scans of the patient in a single procedure, which causes high radiation damage to doctors and patients. This study proposed a large-scale pretrained multi-frame generative model-based real-time and low-dose DSA imaging system (GenDSA). GenDSA was trained and validated by employing millions of DSA images from 35 hospitals, which demonstrated that GenDSA can achieve state-of-the-art performance and reduce the DSA radiation dose to 1/3 of clinically available protocols. Subjective ratings and statistical results from five doctors showed that the generated videos reached a comparable level to the full-sampled videos, both in terms of overall quality and lesion assessment, which fully demonstrated the potential of GenDSA for clinical applications.

SUMMARY

Background: Digital subtraction angiography (DSA) devices are commonly used in numerous interventional procedures across various parts of the body, necessitating multiple scans per procedure, which results in significant radiation exposure for both doctors and patients. Inspired by generative artificial intelligence techniques, this study proposes GenDSA, a large-scale pretrained multi-frame generative model-based real-time and low-dose DSA imaging system.



Methods: GenDSA was developed to generate 1-, 2-, and 3-frame sequences following each real frame. A large-scale dataset comprising ~3 million DSA images from 27,117 patients across 10 hospitals was constructed to pretrain, fine-tune, and validate GenDSA. Two other datasets from 25 hospitals were used for evaluation. Objective evaluations included SSIM and PSNR. Five interventional radiologists independently assessed the quality of the generated frames using the Likert scale and visual Turing test. Scoring consistency among the radiologists was measured using the Kendall coefficient of concordance (W). The Fleiss' kappa values were used for inter-rater agreement analysis for visual Turing tests.

Findings: Using only one-third of the clinical radiation dose, videos generated by GenDSA were perfectly consistent with real videos. Objective evaluations demonstrated that GenDSA's performance (PSNR = 36.83, SSIM = 0.911, generation time = 0.07 s/frame) surpassed state-of-the-art algorithms. Subjective ratings and statistical results from five doctors indicated no significant difference between real and generated videos. Furthermore, the generated videos were comparable to real videos in overall quality (4.905 vs. 4.935) and lesion assessment (4.825 vs. 4.860).

Conclusions: With clear clinical and translational values, the developed GenDSA can significantly reduce radiation damage to both doctors and patients during DSA-guided procedures.

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INTRODUCTION

Image-guided interventional procedures offer high efficiency, minimal trauma, and rapid recovery, becoming a mainstream alternative to more invasive surgical methods.^{1–3} Digital subtraction angiography (DSA) devices, which include scan protocols for fluoroscopy, cone beam computed tomography (CT), and two-dimensional (2D) and 3D angiography imaging, provide physicians with dynamic 2D X-ray images, widely used in interventional procedures (collectively referred to as “DSA imaging”). DSA imaging is extensively utilized for various body parts and organs, covering hundreds of diseases and performed on hundreds of thousands of patients daily worldwide.^{4–9}

However, DSA-guided interventional procedures expose both interventionalists and patients to ionizing radiation. Patients typically receive radiation doses several times or even tens of times higher than routine diagnostic CT scans due to longer acquisition times and more frequent acquisitions.^{10,11} Physicians are also exposed to radiation during DSA-guided procedures as they work alongside patients. This exposure increases the risk of conditions such as arteriosclerosis, leukemia, and cancer, as well as common issues like vision impairment and hair loss.^{12–16} A recent large-scale study tracking the deaths of over 300,000 individuals from France, the United Kingdom, and the United States demonstrated that long-term exposure to low ionizing radiation correlates with a higher risk of cancer deaths. Specifically, for each 1-Gy increase in radiation exposure, the risk of death from cancer rises by 53%, and the risk of death from solid tumors increases by 52%.¹⁷ Another multi-national cohort study following over 940,000 participants who underwent CT exams found a positive association between cumulative radiation dose and the risk of all hematologic malignancies. The study indicated an excess relative risk of 1.96 per 100 mGy.¹⁸ Although radiation doses vary by procedure, a single diagnostic/therapeutic procedure can reach mGy levels. Despite using protective measures, the potential cumulative radiation exposure risk over the years remains significant.

Thus, significantly reducing the radiation dose while maintaining imaging quality to meet clinical needs is a key challenge that DSA imaging has aimed to address since its inception, striving to keep radiation exposure as low as reasonably achievable.^{1,19} DSA imaging protocols can be categorized as follows: (1) scanning different body parts such as the head, thorax, and abdomen; (2) scans targeting blood vessels or non-vessels; and (3) different scanning modes, including stationary devices or rotating devices during continuous exposure. Artificial intelligence (AI)-based methods have been proposed for low-dose DSA imaging.^{20–28} However, some of these methods are only feasible for simulated images, while others are limited to specific scan protocols. There is a clinical need for AI methods that are universally applicable across the diversity of scanning protocols. More critically, none of the algorithms are clinician certified, and most cannot be operated in real time, thus failing to meet clinical practice needs. Consequently, these methods face challenges in clinical translation.

With the rapid development of generative AI technologies such as GPT, Llama, and Sora,^{29–31} generating images and videos based on specific prompts or constraints has become possible. Unlike methods that use text or single images as control conditions, our research focuses on generating complete and stable videos from sparsely sampled video frames. Specifically, we developed a generative model to achieve frame generation for DSA imaging, reducing the frame rate while obtaining real-time intermediate frames that are highly comparable to actual effects.

Challenges in this task primarily arise from complex structures and motions associated with large samples. First, different tissues and vessels exhibit varying structures and complexities, particularly in tiny and irregularly structured tissues, lesions, and blood vessels. Second, the diffusion of the contrast agent involves non-rigid and complex motion, causing significant changes in the signal between consecutive DSA video frames. Third, overlapping movements from patients (e.g., breathing, heartbeat) and machines (e.g., rotation during 3D imaging)

further complicate feature extraction. Despite these challenges, the AI system must efficiently extract these features in real time to effectively assist in clinical procedures.

To tackle the challenges in DSA imaging, GenDSA, a real-time, low-dose imaging system based on a large-scale pretrained multi-frame generative model, was developed. This system generates frames from sparsely sampled DSA videos using a self-supervised pretrained foundation model trained on a diverse and extensive dataset. GenDSA is universally applicable to various DSA scan protocols across different body parts and supports seamless scalability to large datasets without manual annotations. Achieving nearly identical video quality to clinical protocols, GenDSA operates in real time (0.07 s per frame) while reducing the radiation dose by two-thirds.

A large-scale dataset, Digital Subtraction Angiography Videos Dataset of the Multicenter Research in China (DV-MuRC), was created, comprising approximately 3 million DSA images from 27,117 patients across 10 centers. GenDSA was pretrained and fine-tuned on part of this dataset and tested on the remainder, as well as on two additional datasets from 25 medical centers, which included diverse patient demographics and equipment. The novel motion-structure-aware network (MoStNet) algorithm extracts multi-scale complex motion-structural features through five key modules, outperforming the state-of-the-art algorithms extracting motion and appearance (EMA)-video frame interpolation (VFI; computer vision)³² and frame interpolation for large motion (FILM) (streaming used in computer vision).³³ An independent reader study confirmed the clinical relevance of GenDSA by comparing the generated videos with actual clinical videos, demonstrating its translational potential.

RESULTS

Study population

Three datasets were used in this study, with the distribution of patient- and image-related data detailed in Table 1. Participants' information on sex and age was reported by physicians. Information on gender and socioeconomic status was not collected. Dataset 1 included DSA imaging data from 27,117 patients across 10 hospitals in China. Patients underwent various protocols for the head, thorax, abdomen, pelvis, and limbs using GE, Phillips, or Siemens equipment (details on DSA devices are provided in the key resources table and scanning protocols in Table S1). For ease of description, DSA imaging data were categorized based on device motion: (1) "2D-DSA," where images are taken with the machine held at an angle, and (2) "3D-DSA," where images are taken with the machine rotating around the patient for 3D reconstruction. Specifically, the dataset included 22,845 2D-DSA video sequences (~1.05 million images) and 4,272 3D-DSA video sequences (~1.63 million images). For GenDSA pretraining and fine-tuning, 24,240 video sequences (from centers 1–7) were utilized, and 2,877 video sequences (from centers 8–10) were used for external testing, following a 4:4:1 ratio to validate GenDSA's frame generation capability and generalization performance.

To validate the generalization performance of the proposed model across different equipment manufacturers and ethnic populations, two additional datasets were utilized. Dataset 2, named

Data-Neusoft, included cerebrovascular 3D-DSA data from 475 patients at 9 hospitals in China collected by Neusoft Medical Systems. Dataset 3, named Data-Netherlands, comprised 2D cerebrovascular DSA images from 2,396 patients in the MR CLEAN Registry³⁴ collected from 16 comprehensive stroke hospitals in the Netherlands. The entire study and its description adhered closely to the Checklist for Artificial Intelligence in Medical Imaging,^{35,36} which is the most commonly used standard in the field.

A unified framework for GenDSA

Figure 1 illustrates the clinical application strategy and framework of GenDSA. Figure 1A provides an overview, showing that GenDSA can acquire videos almost identical to clinically available protocols in real time (0.07 s/frame) using only one-third of the radiation dose. The dashed line represents the operation mechanism of GenDSA. When the temporal resolution of DSA scanning is undersampled to one-third, GenDSA generates two virtual frames (F_{n+1} , F_{n+2}) that are highly comparable to the actual results after each captured frame (F_n). GenDSA seamlessly interfaces with DSA equipment, and Figure S1 demonstrates the results of our body-mimicking experiments simulating clinical reality. During the DSA scanning process, GenDSA-generated images can be projected in real time to the operating room for doctors' reference (Figure S1). Figure 1B schematically represents GenDSA. Partial 2D data covering five body parts (head, thorax, abdomen, pelvis, and limbs) were used for pretraining and input into the proposed MoStNet to obtain the pretrained foundation model. Additionally, the model was fine-tuned separately with 2D/3D data from each part. Data from each body part were also used for fine-tuning, with the specific amounts detailed in Table 1. MoStNet, the core algorithm of GenDSA, addresses the real-time generation of DSA images. Figure 1C illustrates the concise network architecture of MoStNet. First, multi-scale features are extracted from input frames and enhanced through cross-scale fusion. The motion-structure feature extractor (MSFE) module extracts motion and structural features between enhanced frames using cross-attention. Unlike Transformer models, which suffer from high computational complexity, MSFE calculates cross-attention in a fully convolutional manner, converting contexts into linear functions to efficiently extract features and reduce computational complexity. This approach significantly reduces image generation time (Figure 2C), meeting the demand for real-time DSA imaging in clinical procedures. Next, motion features are mapped at different times and decoded with structural features to compute flow and masks. Finally, a simplified UNet refines features at different scales, decoding the flow, masks, and structural features to produce the corresponding intermediate frame I_t . Detailed descriptions of GenDSA and MoStNet are provided in the STAR Methods and Figure S2. Since the data used in this study were collected retrospectively, specific dose values for low-frame-rate acquisition in clinical situations were unavailable. To estimate theoretical radiation dose savings, 700 patients across seven different tasks were randomly selected, their reports were reviewed, and the radiation dose received during a single DSA scan was counted. Theoretical radiation dose savings were estimated based on multiples of the actual sparing (Table S2).

Table 1. Patient- and DSA-image-related baseline information

	2D			3D	
	Pretraining	Fine-tuning	Testing	Fine-tuning	Testing
Dataset 1: 27,117 video sequences					
Age (available data)					
Median (years)	56	55	54	59	58
IQR (years)	16	13	11	16	19
Age bucket					
<18, <i>n</i> (%)	88 (0.75)	56 (0.74)	17 (0.78)	22 (0.57)	3 (0.73)
18–30, <i>n</i> (%)	404 (3.40)	257 (3.40)	75 (3.46)	99 (2.58)	15 (3.66)
31–60, <i>n</i> (%)	6,184 (52.05)	3,935 (52.05)	1,125 (52.00)	1,923 (50.16)	205 (50.00)
>60, <i>n</i> (%)	5,204 (43.80)	3,312 (43.81)	947 (43.76)	1,790 (46.69)	187 (45.61)
Gender (available data)					
Male, <i>n</i> (%)	7,583 (63.56)	4,829 (63.58)	1,386 (63.64)	2,286 (59.63)	238 (54.34)
Female, <i>n</i> (%)	4,347 (36.44)	2,766 (36.42)	792 (36.36)	1,548 (40.37)	200 (45.66)
Body parts					
Head and neck	3,697 (30.23)	2,550 (31.29)	735 (29.79)	626 (16.32)	213 (48.63)
Thorax	733 (6.00)	570 (6.99)	185 (7.50)	2,381 (62.10)	121 (27.62)
Abdomen	5,394 (44.11)	3,455 (42.39)	1,024 (41.51)	730 (19.04)	76 (17.35)
Pelvis	1,354 (11.07)	884 (10.85)	288 (11.67)	97 (2.54)	24 (5.50)
Limbs	1,050 (8.59)	691 (8.48)	235 (9.53)	0 (0.00)	4 (0.90)
Multiple institutions					
Center 1	8,879 (72.61)	5919 (72.62)	–	2,240 (58.42)	–
Center 2	1,634 (13.36)	1,089 (13.36)	–	1,269 (33.10)	–
Center 3	566 (4.63)	377 (4.62)	–	36 (0.94)	–
Center 4	502 (4.11)	334 (4.10)	–	162 (4.23)	–
Center 5	309 (2.53)	206 (2.52)	–	75 (1.95)	–
Center 6	285 (2.33)	190 (2.36)	–	52 (1.36)	–
Center 7	53 (0.43)	35 (0.42)	–	–	–
Center 8	–	–	1,270 (51.48)	–	369 (84.40)
Center 9	–	–	309 (12.52)	–	69 (15.60)
Center 10	–	–	888 (36.00)	–	–
Dataset 2: 475 video sequences					
Head	–	–	–	–	475 (100.00)
Dataset 3: 2,396 video sequences					
Head	–	–	2,396 (100.00)	–	–

Quantitative evaluation of generated DSA videos

To maintain video quality while reducing the frame rate, the optimal number of generation frames was investigated based on sequences of 1-, 2-, and 3-frame generations after each real frame was captured. These strategies can reduce the radiation dose to 1/2, 1/3, and 1/4 of current clinical protocols, referred to as "generated 1 frame," "generated 2 frames," and "generated 3 frames," respectively. According to the experimental results of [Data S1](#) shown in [Figure 2A](#), there were no severe drops in peak signal-to-noise ratio (PSNR) or structural similarity metrics (SSIMs) with 1-, 2-, and 3-frame generations and no significant increase in generation time per frame. The mean $\pm$ SD PSNR and SSIM values for 1-, 2-, and 3-frame generations across all body parts (both 2D and 3D) were 37.60 ± 4.58 , 36.83 ± 4.48 , and 36.15 ± 4.67 and 0.916 ± 0.048 , 0.911 ± 0.050 , and $0.900 \pm$

0.055 , respectively ([Figures S3A and S3B](#)). The mean generation times for 1-, 2-, and 3-frame generations across all parts were 0.057, 0.146, and 0.236 s, respectively ([Figure S3C](#)). The objective scoring (PSNR and SSIM) of [Data S1](#) demonstrated excellent performance of GenDSA for both overall and specific body parts. [Figure 2B](#) shows a key frame of a 3D head from a randomized patient, consistent with the effect of the image matching the objective scores. Furthermore, comparing the residual plots of the 1- and 2-frame generation images with fully sampled frames showed no significant residual signals, while the 3-frame generation images had weak residual signals. Critically, the 3-frame generation videos exhibited inferior quality compared to the other schemes, evidenced by compromised video clarity and fluency (source file 1 within [Data S1](#)). Therefore, the 2-frame generation model was deemed the most suitable for practical clinical

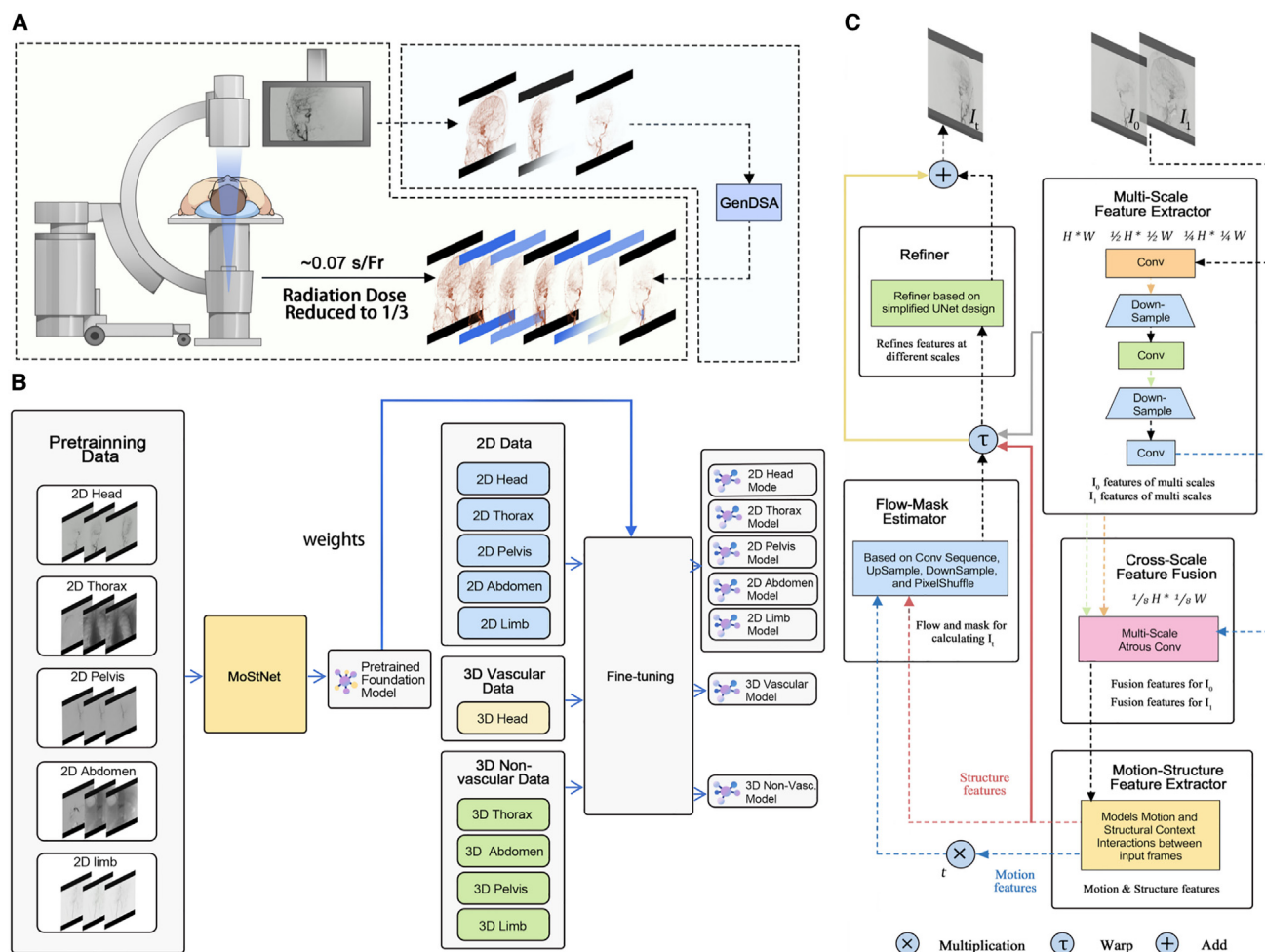


Figure 1. Overview of the GenDSA process and network architecture

(A) An illustration of DSA shooting process with GenDSA. The solid line represents the part experienced by doctors and patients, while the dotted line represents the backend processing process.

(B) The schematic representation of the proposed GenDSA.

(C) Overall network pipeline of MoStNet.

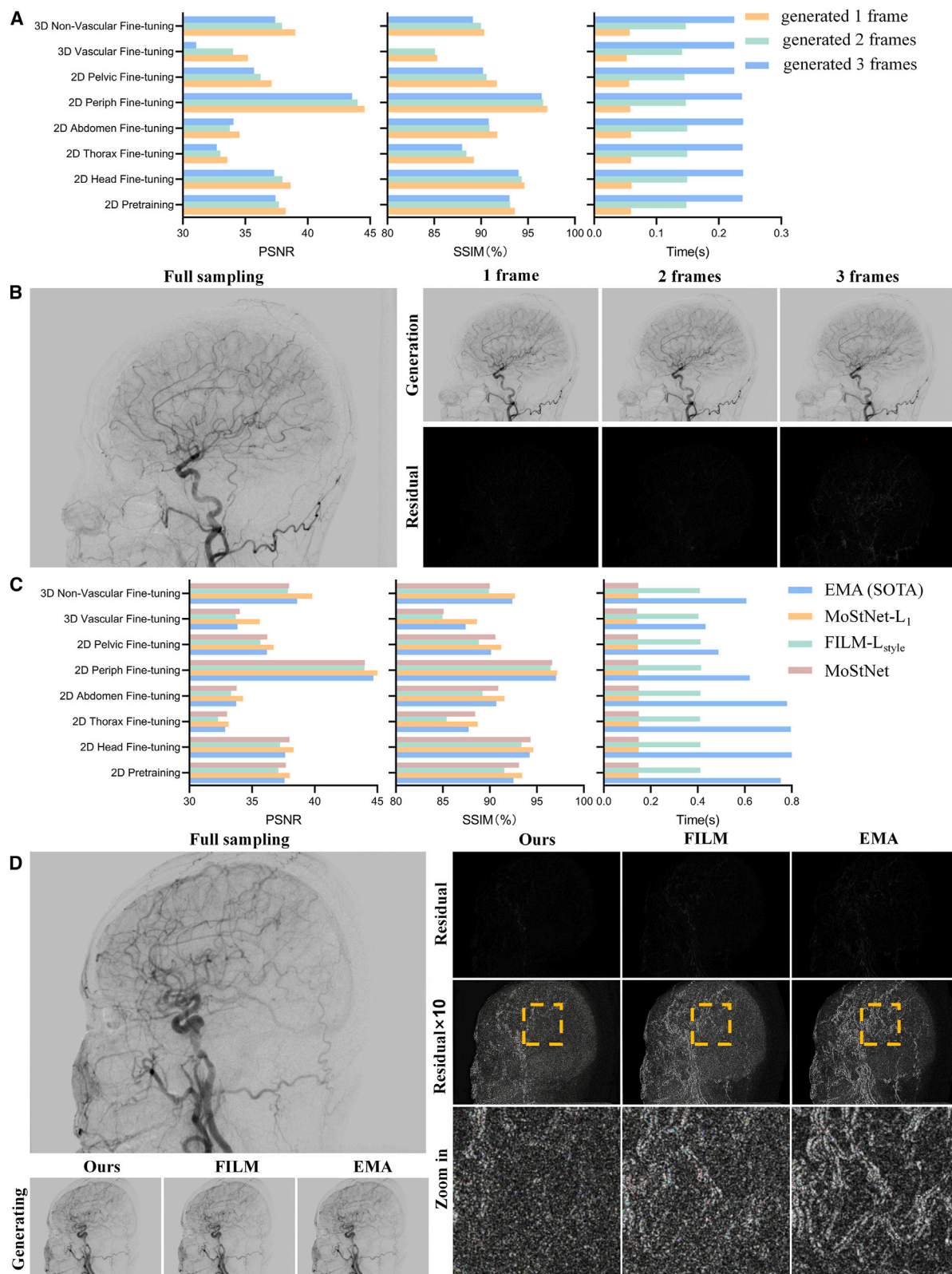
applications, providing high-quality videos while minimizing radiation dose.

GenDSA was compared to state-of-the-art EMA and streaming-used FILM algorithms for 2-frame generation, with the PSNR and SSIM metrics detailed in Figure 2C. To ensure a fair comparison, MoStNet, GenDSA's core algorithm, employed two distinct loss functions: L1 (MoStNet-L1) and a combined loss function (MoStNet). Higher PSNR and SSIM metrics were achieved with the L1 loss function, although it resulted in worse visual effects (see the ablation experiment section for details). Figure 2C shows MoStNet-L1 outperforming both peer algorithms across all metrics, while MoStNet surpassed FILM on all metrics and EMA on several metrics (mean PSNR, SSIM, and generation times are presented in Figures S3D–S3F). Figure 2D compares MoStNet, FILM, and EMA using a key frame of a 3D head from a random patient. Residual plots indicate slightly more noticeable residuals in EMA-generated images, especially

with 10 $\times$ zoom and enhancement. EMA exhibited limitations in video fluency, particularly for 3D-DSA, as seen in Videos S1 and S2 within Data S1, which show significant left-right oscillations in EMA-generated videos. In contrast, MoStNet-generated videos did not exhibit these issues. Furthermore, MoStNet improved frame generation speed to 0.147 s on average, 4.4 and 2.7 times faster than EMA (0.661s) and FILM (0.410s), respectively. This speed is crucial for real-time imaging during interventions, demonstrating MoStNet's suitability for this task.

Evaluation of multi-scenario utilization

The capability of GenDSA to generate video sequences for patients with multi-site disease is shown in Figure 3. The generated projections precisely reconstruct intracranial large and small vessels, revealing the dynamic filling process of cerebral blood flow. This is evidenced by the nearly non-existent signal on the key frame of the residual projection in Figure 3A. Three random



(legend on next page)

pixels were annotated at distal and proximal vessels to plot time-intensity curves on fully sampled, undersampled, and generated videos. Compared to undersampled videos, the generated videos exhibited smoother curves, more comparable to those of fully sampled projections (Figure 3E). The complete frame sequence of the case in Figure 3A is dynamically presented in Video S3 within Data S1, corroborating the description. As shown in Video S3, time-intensity changes may indicate pathological states and/or hemodynamic information. Our results demonstrated that the signal intensity errors with the generated videos were maintained below 0.02 when normalized to [0,1], while undersampled videos exhibited errors exceeding 0.1 during sharp signal changes, more than five times higher. Figure 3B shows that the generated sequence for small vessels in a 2D thorax image conformed to the fully sampled video sequence. For key frames with aneurysms and ablation needles, rotational 3D-DSA images generated by GenDSA matched the original images well (Figures 3C and 3D). Notably, the displacement of the needle tip between the generated and fully sampled videos was within 2 pixels, significantly lower than that in undersampled videos (Figure 3F). The complete frame sequence of the case in Figures 3D and 3F is dynamically displayed in Video S4 within Data S1. As shown in Video S4, the proposed method accurately guides the percutaneous puncture procedure while reducing radiation doses for physicians and patients to one-third of current protocols. However, without the proposed approach, larger errors (up to 6 pixels, ~3 mm) were observed when the frame rate was reduced to one-third. Overall, examples of video sequences for all 2D and 3D parts were randomly selected for illustration, and the corresponding video sequences are shown in source file 2 within Data S1. Key frames and videos inferred by GenDSA with 1- and 2-frame generations matched fully sampled ones, demonstrating satisfactory reproduction of the movement of the puncture needle tip, cerebral blood vessels, and stenosis sites.

Clinical assessment

Five interventional radiologists with 8–15 years of expertise blindly assessed 474 pairs of actual (fully sampled) and generated DSA videos from 9 different body parts using 2D/3D scanning methods. The 474 pairs included 50 pairs each of 2D head, 2D thorax, 2D abdomen, 2D pelvis, and 2D limb; 100 pairs of 3D head; 50 pairs each of 3D thorax and 3D abdomen; and 24 pairs of 3D pelvis-limbs. This study evaluated the proposed GenDSA from two key dimensions.

Assessment of the capability among interventional radiologists in differentiating generated videos from real videos

Interventional radiologists were asked to dichotomize shuffled videos generated by GenDSA and real clinical videos. According

to the confusion matrix for 2D- and 3D-DSA images, none of the five radiologists could accurately differentiate between actual and generated videos, regardless of whether 1-frame or 2-frame generation was used (Figures 4A and S4). Significantly poorer agreement (k values ranging from 0.040 to 0.168, all $p < 0.05$) was observed among the readers in identifying generated videos from actual ones, demonstrating that the GenDSA-generated videos were indistinguishable from real videos (Table S3).

Evaluations of overall performance and multi-site lesion diagnosis capacity of GenDSA

The overall performance and lesion diagnosis capacity of GenDSA were evaluated on fully sampled, generated, and undersampled videos, with scoring criteria detailed in the STAR Methods section. For 2-frame generation, generated videos were noticeably superior to undersampled videos by 5-level image quality scores (Figure 4B). Generally, the overall performance and multi-site lesion diagnosis of generated videos (4.905 and 4.825 points) were comparable to fully sampled videos (4.935 and 4.86 points) and significantly superior to undersampled videos (3.28 and 3.63 points). The differences between generated and fully sampled videos for 2D and 3D overall performance and 2D and 3D multi-site lesion diagnosis were 0.05, 0.01, 0.05, and 0.02 points, respectively (all $p > 0.05$, Tables S4 and S5). The average rating results for all body parts exceeded 4.6 points, indicating that the generated videos did not compromise physicians' diagnosis and treatment outcomes. Moderate consistency was observed among the classification results of the generated videos, with Kendall's W values ranging from 0.565 to 0.791 (all $p < 0.001$, Tables S6B and S6D), comparable to fully sampled videos and significantly better than undersampled videos.

The generated 2D- and 3D-DSA videos with 1-frame generation demonstrated significantly superior lesion identification capacity, overall quality, and fluency compared to undersampled videos and slightly outperformed 2-frame generation videos (Figure S5; Tables S4, S5, S6A, and S6C).

External validation

To verify GenDSA's generalizability across various countries and manufacturers, two external datasets from the Netherlands and Neusoft Medical Systems were utilized. Due to the significant differences between the training data and these test datasets, the 2-frame generation results were unsatisfactory. Therefore, only the 1-frame generation results are presented, which effectively reduces the radiation dose to half of the clinically available protocols. For 3D vascular data, the critical credential for assisting doctors in diagnosis and treatment is the 3D-reconstructed images. Therefore, the advantages of our frame generation models for 3D reconstruction were further evaluated using clinically

Figure 2. Comparative analysis of image generation metrics and techniques in GenDSA models

(A) Comparisons of PSNR, SSIM, and time for generations among the three generated schemes for 7 fine-tuning models (i.e., 2 models for 3D-DSA and 5 models for 2D-DSA) and one pretraining model.

(B) 1-, 2-, and 3-frame generation results and the associated residual images are shown.

(C) Comparisons of PSNR, SSIM, and time for generations for the 2-frame generation scheme among EMA, MoStNet-L₁, FILM-L_{style}, and MoStNet for 7 fine-tuning models and one pretraining model.

(D) Comparisons of the generated images and residual images based on the 2-frame generation scheme by using EMA, FILM, and MoStNet.

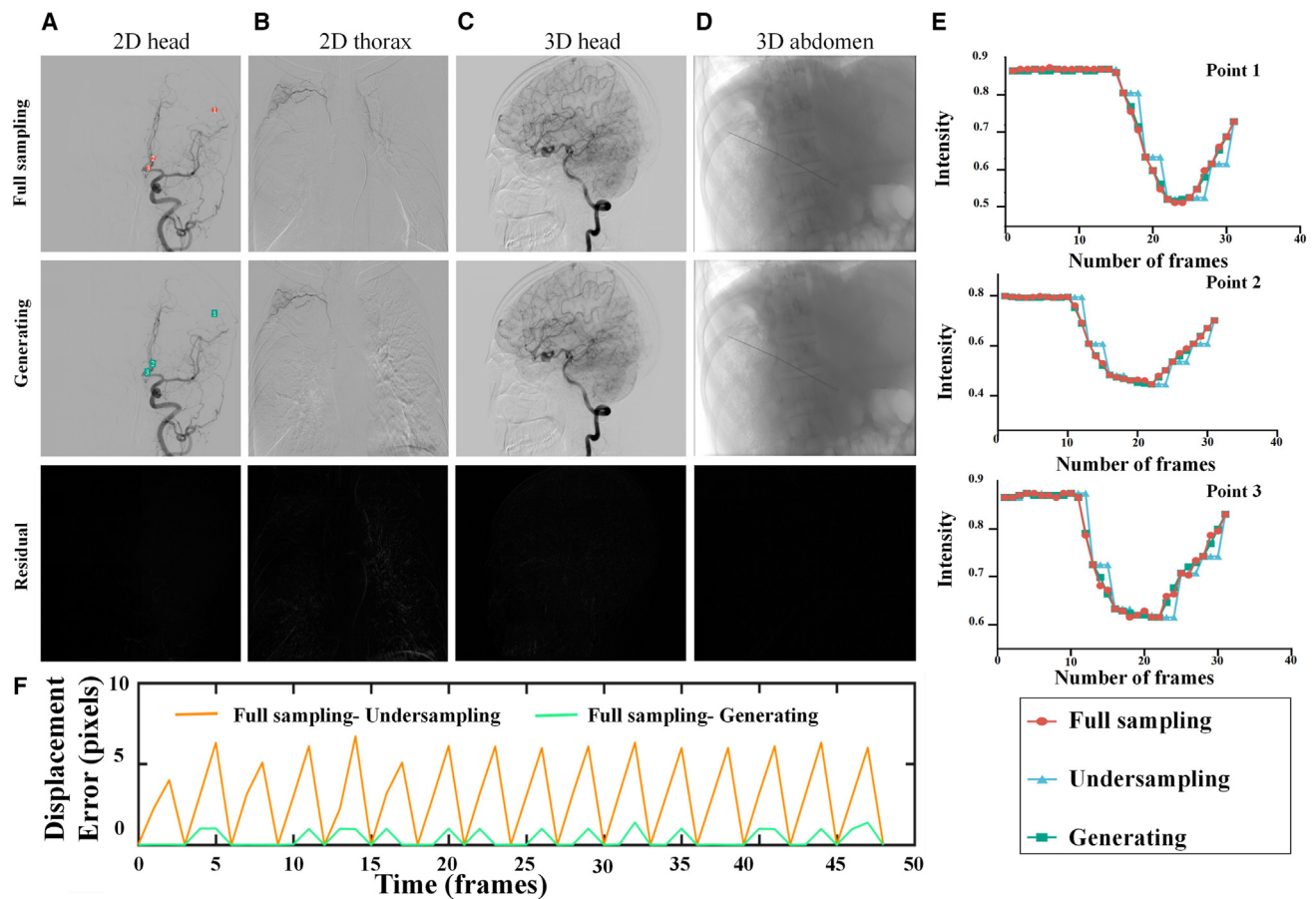


Figure 3. Results of GenDSA with 2-frame generation

(A and B) The sample frames of generated 2D-DSA projections for head and thorax imaging.

(C and D) Key frames of generated 3D-DSA projections for intracranial aneurysms and ablation needle imaging.

(E) The time-intensity curves for the fully sampled, undersampled, and generated videos corresponding to vessel pixels illustrated in (A).

(F) The displacement curve of the needle point on frame undersampled and generated videos relative to the fully sampled videos.

available software to reconstruct 3D images from fully sampled, undersampled, and generated frames. Video S5 within [Data S1](#) shows the reconstructed image effects from multiple viewpoints and different scales, revealing that the undersampled frame reconstructions were extremely poor, with clearly visible noise throughout the image. In contrast, the reconstruction results using our frame generation approach were highly comparable to the fully sampled ones. [Figures 5A](#) and [5B](#) demonstrate superior performance over the undersampled approach based on a patient with internal carotid artery stenosis. Two key frames from Video S5, at 3 and 13 s, illustrate comparable 3D-reconstructed results from fully sampled, undersampled, and generated frames. The enlarged field of view of localized vasculature in [Figure 5B](#) shows that the 3D-reconstructed results from generated frames completely preserved the morphology and structure of cerebral vessels. [Figure 5B](#) also shows that the measured stenosis vessel diameter of 1.6 mm was in good agreement with the fully sampled frames' 3D-reconstructed results, while the undersampled frames introduced a measurement error (1.3 instead of 1.6 mm). Additionally, [Figure 5C](#) presents the key frame of the

2D-DSA video sequences for a subject (Data-Netherlands) with no significant cerebral vascular abnormalities. A visual comparison indicates that the generated frame is comparable to the fully sampled frame, with no discernible signals in the residual image. Encouragingly, the proposed GenDSA produced excellent 2D- and 3D-DSA frames even when the input images were collected from different countries and manufacturers, as evidenced by the satisfactory PSNR and SSIM values achieved: 38.94 ± 3.38 and 0.935 ± 0.046 (Data-Neusoft) and 34.51 ± 3.92 and 0.941 ± 0.028 (Data-Netherlands) ([Figure 5D](#)). Overall, most SSIM and PSNR metrics exceeded 0.85 and 30, proving a high degree of consistency between the generated and fully sampled videos.

Ablation experiment

To evaluate feature importance in the proposed approach, ablation experiments were conducted. The impact of context modeling scope (r) changes on the generated results was assessed. [Figure 6](#) demonstrates that after fine-tuning, the impact of r on mean and SD is minimal, with mean/SD differences in SSIM and PSNR within 0.002/0.004 and 0.6/0.3, respectively,

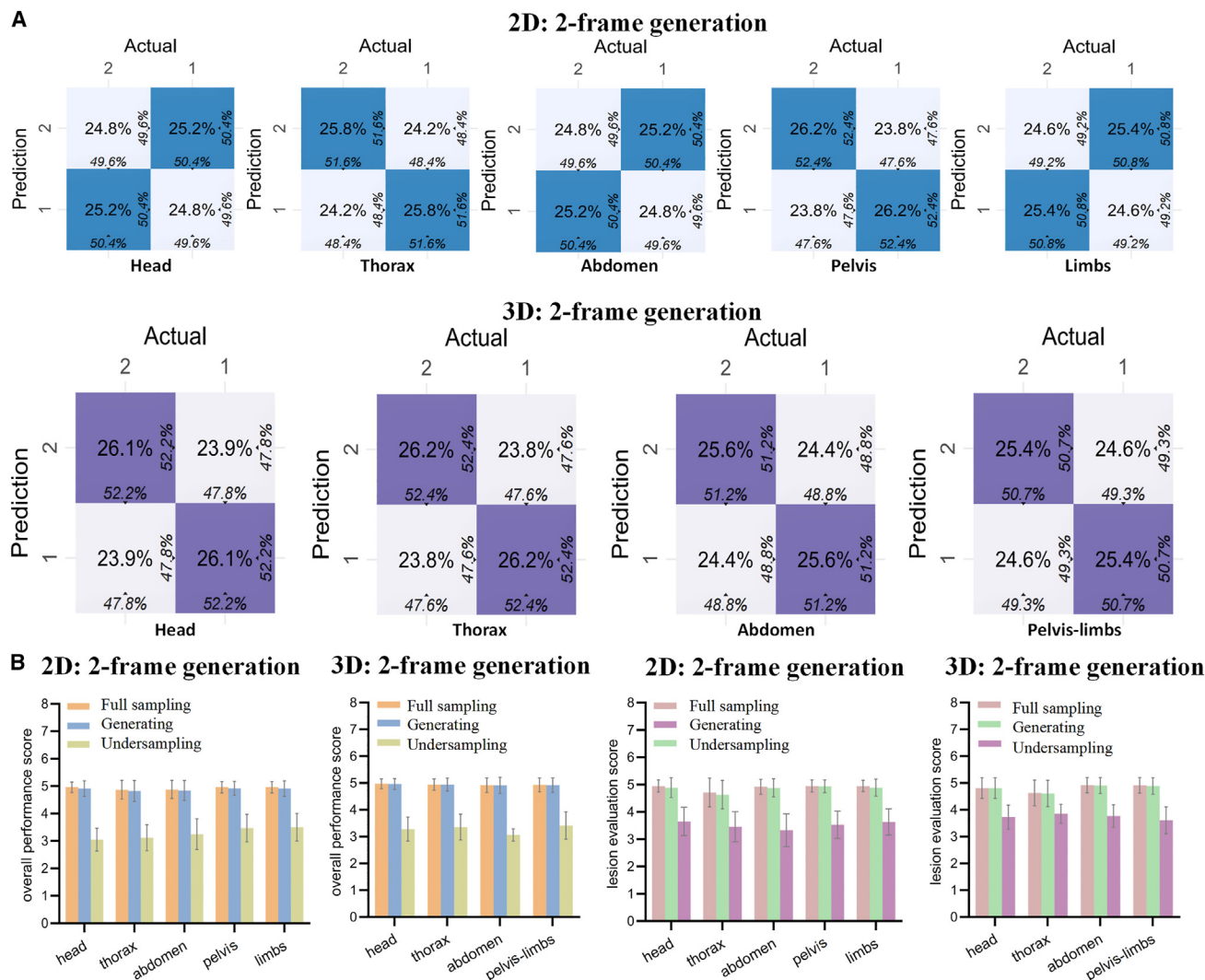


Figure 4. Clinical assessments of 2-frame-generation videos using GenDSA

(A) Confusion matrices of discriminating actual 2D- and 3D-DSA videos from 2-frame generation videos.

(B) Rating results of 5 interventional radiologists by 5-level image quality scores.

highlighting the robustness of the MSFE module. Additionally, comparisons of two loss functions were implemented based on the proposed MoStNet. Figure S6 shows that the proposed loss function significantly improved image quality. Specifically, the sole application of L1 resulted in blurred image details (blue box), whereas the combined loss function (green box) not only eliminated blurriness but also closely approximated the ground truth (red box).

DISCUSSION

In summary, to significantly reduce the DSA radiation dose through frame generation, this study proposed a generative model-based AI system (GenDSA). A large dataset (DV-MuRC) containing approximately 3 million DSA images from 10 hospitals was created to pretrain, fine-tune, and externally test the model,

enhancing and validating GenDSA's generalization performance. Additionally, two external datasets from 25 medical centers, comprising patients of various ethnicities and data acquired with instruments from different manufacturers, were used to further validate the generative ability and generalization performance of GenDSA. The system's performance was assessed through subjective and objective evaluations on large and diverse datasets. In conclusion, this study validates GenDSA as the only AI method currently universally applicable to various scanning protocols for ultra-low-dose DSA imaging that meets clinical needs, evaluated on three key dimensions: imaging time (crucial for real-time DSA-guided procedures), imaging quality, and generalization performance.

VFI techniques, commonly used in computer vision to enhance temporal resolution,^{37–42} face challenges in medical imaging due to the distinct characteristics of captured objects from

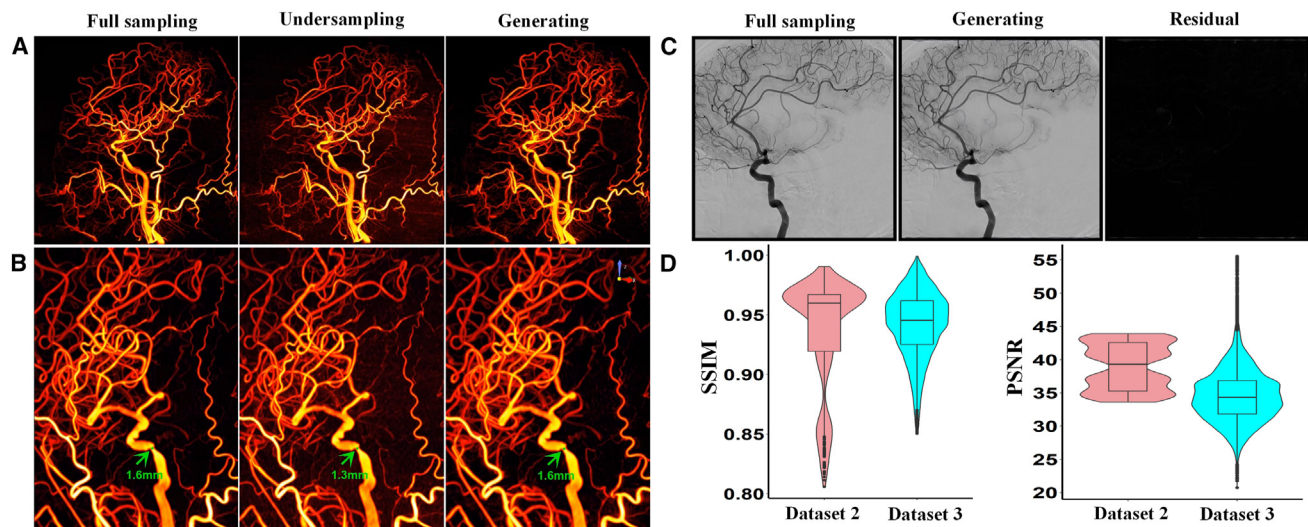


Figure 5. Results of GenDSA with 1-frame generation on external datasets

(A) Visual comparison of 3D reconstruction results of the fully sampled, undersampled, and generated frames from 3D-DSA images from a patient with intracranial artery stenosis.

(B) Locally magnified views of (A) that were marked with the diameter of stenosis vessel.

(C) The key frame was chosen from a 2D-DSA video of a subject without significant abnormalities, showing ground truth (left), the generated image (middle), and the residual image (right).

(D) Quantitative evaluation of the similarity between generated videos obtained from 2 datasets via violin plots.

natural videos. These limitations hinder the application of algorithms like EMA³² and FILM³³ in medical contexts. This study demonstrates that even advanced methods fail to generate DSA frames that meet clinical needs despite extensive pretraining and fine-tuning. No pretraining algorithm currently addresses frame generation for DSA or other prevalent medical imaging modalities. To tackle this, MoStNet, incorporating the MSFE module, was proposed. MSFE efficiently extracts complex motion and structural features at multiple scales via cross-attention in a fully convolutional manner, forming the core of MoStNet and, thus, GenDSA. MoStNet surpasses EMA and FILM in generating low-dose DSA images through frame generations. EMA (0.661 s per 2 frames) and FILM (0.410 s per 2 frames) fall short of real-time intervention needs. EMA's inability to extract complex vascular structures and motions compromises video fluency, particularly for 3D-DSA. MoStNet's intricate network architecture addresses these issues, eliminating video jumping and vascular beating. Additionally, MoStNet produces fewer residual signals than FILM and EMA, resulting in more accurate generated videos.

When fine-tuning the model for n -frame interpolation, we trained a dedicated model for each of the five 2D regions, with distinct models for the 3D vascular and 3D non-vascular categories. While this approach is advantageous for specific clinical applications, it also results in a large number of models lacking generalizability. In future research, we plan to develop more generalized models capable of fulfilling the diverse clinical application needs across various regions with a single model. Notably, while the 3-frame generation can reduce the radiation dose to one-fourth of the regular dose, it yields inferior video clarity and fluency compared to the 1-frame and 2-frame gener-

ation schemes. This may be due to the overly undersampled DSA images, which miss information that cannot be generated by the proposed approach. However, a reduction of approximately 67% in radiation dose is achievable with the 2-frame generation, greatly benefiting both physicians and patients. Despite varying levels of proficiency with different anatomical procedures among the five interventional radiologists, their diagnostic ability remained unaffected. Diagnosing DSA images was straightforward for radiologists with over 8 years of experience. The reliability of their scores was confirmed by the high degree of consistency in inter-rater agreement analysis of the overall image performance and lesion diagnosis.

Lastly, the overall generalizability of GenDSA was demonstrated using two additional datasets. Based on Data-Neusoft, GenDSA provided a more accurate measure of stenotic vessel diameter than approaches without frame generation, highlighting its clinical value. Additionally, the absence of residual signals in the residual image based on Data-Netherlands further corroborated its generalizability. The reliability of GenDSA was also supported by two quantitative metrics, PSNR and SSIM. These results firmly establish the generalizability of GenDSA for clinical applications.

DSA-guided surgery has been widely adopted by surgeons in orthopedics, neurosurgery, hepatobiliary, cardiac, and thoracic surgery departments, and its use continues to grow. Recently introduced clinical devices equipped with low-dose DSA imaging were demonstrated to achieve reduced radiation by lowering the dose per frame.^{43–46} In contrast, the present study reduces radiation by decreasing the frame rate. A combination of our technique with low-dose DSA imaging devices could further reduce radiation exposure. Our AI system can be integrated by

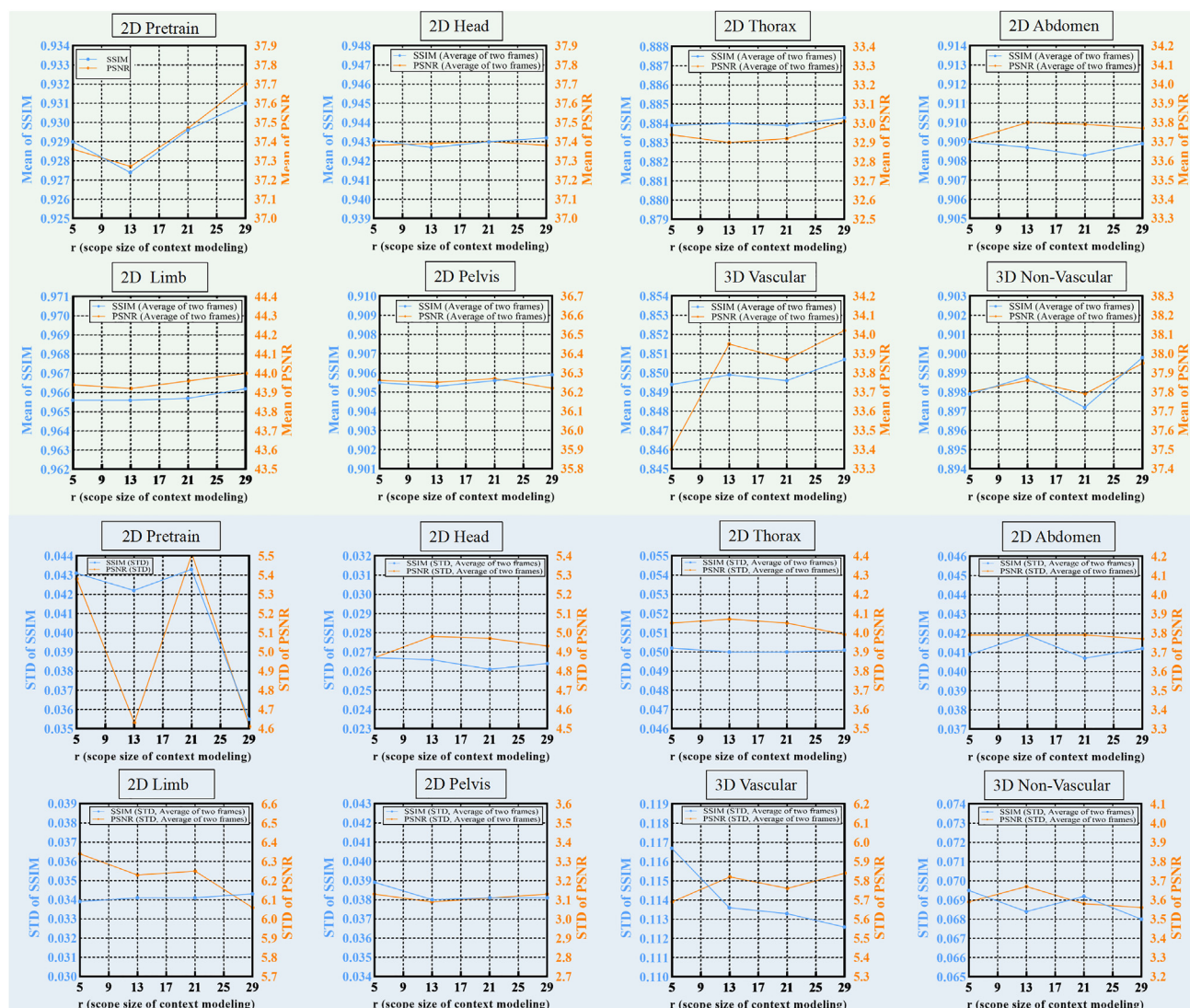


Figure 6. |Results of ablation experiment

Assessment of the impact of context modeling scope (r) changes on PSNR, SSIM, and SD of the generated videos with respect to the 7 fine-tuned models (i.e., 2 models for 3D and 5 models for 2D) and one pretraining model.

various DSA device manufacturers or developed as a standalone system for downstream tasks in DSA imaging; this would significantly reduce radiation exposure for doctors and patients.

Limitations of study

The limitations of this study include the need for further verification of GenDSA's capabilities in more clinical trials despite the validation of its clinical translational value from multiple aspects. Additionally, while our sample included dozens of diseases across various body parts, we did not categorize these diseases in detail for clinical assessment. As DSA imaging has been used clinically for hundreds of diseases, one scanning protocol alone can be used for up to dozens of diseases. The sample size Data S1 in data set 1 (27,717 patients with millions of images) is already very large in the field of medical imaging, even though

the 2,000 cases of test data we used are still insufficient when allocated to each disease, so we did not further analyze by disease type but only classified according to the main scanning protocols. The DV-MuRC dataset is growing by approximately 500 patients per month. When the incremental data volume reaches a sufficient scale, future clinical studies will categorize the diseases in more detail.

STAR★METHODS

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SUPPLEMENTAL INFORMATION

Supplemental information can be found online at <https://doi.org/10.1016/j.medj.2024.07.025>.

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AUTHOR CONTRIBUTIONS

Conceptualization, H.Z., X.W., and C.Z.; data collection, L.W., L.C., J.M., W.Z., T.S., Y. Lei, X.H., T.N., N.W., H.H., Y.T., W. Lu, W.Y., K.L., R.S., D.X., G.L., and Y.H.; methodology, H.Z., Z.X., Z.C., and X.W.; investigation, H.Z., X.W., and C.Z.; visualization, H.Z., L.W., Z.X., and Y.F.; funding acquisition, H.Z. and C.Z.; project administration, H.Z., X.W., and C.Z.; supervision, H.Z., X.W., and C.Z.; writing – original draft, H.Z., X.W., Z.X., X.L., Y. Li, C.F., W. Liu, X.K., G.T., R.S., P.M.v.d.S., C.B.L.M.M., W.H.v.Z., A.v.d.L., and T.v.W.; writing – review & editing, all authors. H.Z. and C.Z. had unrestricted access to all data. All authors read and approved the final article and take responsibility for its content.

DECLARATION OF INTERESTS

The authors declare no competing interests.

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STAR★METHODS

KEY RESOURCES TABLE

REAGENT or RESOURCE	SOURCE	IDENTIFIER
Software and algorithms		
Python 3.8	Python	https://www.python.org
PyTorch 1.9.0	PyTorch	https://pytorch.org
MATLAB R2018a	MATLAB	https://ww2.mathworks.cn
Imaris 10.0	Imaris	https://imaris.oxinst.com/
R 4.3.1	R	https://www.r-project.org/
GenDSA code	This paper	https://github.com/ZyoungXu/GenDSA
Deposited data		
DV-MuRC	Zhao et al.	This paper
MR CLEAN Registry	Jansen et al. ³⁴	https://doi.org/10.1136/bmj.k949
Data-Neusoft	Zhao et al.	This paper
Other		
DSA equipment 1	Philips Medical Systems	Allura UNIQ FD20
DSA equipment 2	Philips Medical Systems	Allura Xper FD20
DSA equipment 3	Philips Medical Systems	Azurion 7 M20
DSA equipment 4	Philips Medical Systems	Azurion 7 C20
DSA equipment 5	Siemens Medical Systems	Artis zee III ceiling
DSA equipment 6	Siemens Medical Systems	Somatom Confidence
DSA equipment 7	Siemens Medical Systems	Artis zee floor
DSA equipment 8	Siemens Medical Systems	ARTIS pheno
DSA equipment 9	General Electric Healthcare	Innova IGS 540
DSA equipment 10	General Electric Healthcare	Discovery IGS 730
DSA equipment 11	General Electric Healthcare	Optima IGS 330

RESOURCE AVAILABILITY

Lead contact

Further information and requests for resources should be directed to and will be fulfilled by the lead contact, Huangxuan Zhao (zhao_huangxuan@sina.com).

Materials availability

This study did not generate new unique reagents.

Data and code availability

The data presented in the figures of [Data S1](#) are all publicly available in the data source. Cerebrovascular 3D-DSA data from 50 cases and chest CBCT data from another 50 cases in Dataset 1 are publicly available (<https://github.com/ZyoungXu/GenDSA>) for research use, the remaining data of Dataset 1 is available for research purposes upon reasonable request from the corresponding author, Huangxuan Zhao. The rest of the data are not publicly available owing to restrictions in the data-sharing agreement. The source code, pretrained, and fine-tune models of the algorithm is available for research uses at <https://github.com/ZyoungXu/GenDSA>.

EXPERIMENTAL MODEL AND STUDY PARTICIPANT DETAILS

Overall description of the scientific research problem and solution

DSA imaging causes radiation damage to both the physician and the patient and multiple scans are required for a single procedure, resulting in the use of a large amount of radiation. Image generation techniques can efficiently reduce the radiation dose, but such an image generation technique for DSA sequences has not been available yet due to the complexity of motion and tissue/vessel structure. To solve such a problem, we proposed a large-scale pretrained multi-frame generative model-based real-time and low-dose

DSA imaging system (GenDSA). Specifically, if we need to acquire $3n$ frames to form a DSA video sequence, the corresponding frames are named $F_1, F_2, F_3, \dots, F_{n-2}, F_{n-1}, F_n, \dots, F_{3n-2}, F_{3n-1}, F_{3n}$. With GenDSA, we can reduce the temporal resolution to $1/3$ of the previous one, i.e., we only need to capture the frames $F_1, F_4, \dots, F_n, \dots, F_{3n}$ and generate intermediate frames with GenDSA.

Study design

In order to fully train and validate GenDSA, pretraining, fine-tuning and testing were conducted with 3 datasets from 35 centers in this retrospective study. To test the generative results under different temporal sparsity sampling schemes, $1/2, 1/3, 1/4$ sparse sampling data were used as the inputs. Specifically, three different schemes of the GenDSA model were realized by generating one frame, two frames, and three frames after each real frame, which were referred to as “1-frame generation”, “2-frame generation” and “3-frame generation”, respectively, so that “the full-sample similar video” can be obtained. For the purpose of evaluating the performance and generalization capabilities of the GenDSA model, fully sampled data (i.e., using the current clinical protocols) were employed as the gold standard, and PSNR/SSIM values and generation time were utilized as the objective quantitative indicators. In addition, the scores recorded by the radiologists were used as the subjective indicators to assess whether the frames generated by GenDSA meet the criteria for clinical use. This retrospective study was reviewed and approved by the institutional review boards of the Medical Ethics Committee of the corresponding hospitals.

List of DV-MuRC hospitals

Wuhan Union Hospital (center 1), Zhongda Hospital (center 2), Hubei Provincial Hospital of Integrated Chinese and Western Medicine (center 3), Renmin Hospital of Wuhan University, Hubei General Hospital (center 4), Zhongnan Hospital of Wuhan University (center 5), Wuhan Third Hospital, Tongren Hospital of Wuhan University (center 6), Hubei Material and Child Health Hospital (center 7), Jinyin Lake Branch, Wuhan Union Hospital (center 8), West Branch, Wuhan Union Hospital (center 9), Wuhan Tongji Hospital (center 10).

A comprehensive description of the MoStNet

The advanced MoStNet provided a novel method to guide the multi-frame generation of DSA images. Figure S2 depicts the overall network architecture. The network architecture particularly constitutes five key modules, including the Multi-Scale Feature Extractor (FE), the Cross-Scale Feature Fusion (CSFF), the Motion-Structure Feature Extractor (MSFE) modules, the Flow-Mask Estimator (FME), and the Refiner modules. FE, CSFF, and MSFE were instrumental in the extraction of the general motion and structural features. In contrast, the Refiner and the FME modules were used to decode and refine these features for various moments t to produce the frame I_t .

METHOD DETAILS

General motion and structural features extraction

The Multi-Scale Feature Extractor (FE) module

The FE module was employed to ascertain the underlying features of blood vessels of distinct sizes prior to extracting motion and structural features and to derive neurovascular features at three distinct scales. The 3×3 convolution was executed on the input frames I_0 and I_1 , facilitating the calculation of the third layer of low-level features G_0^0 , and G_1^0 . Subsequently, a PReLU activation function was adopted. Then, through the aforementioned convolution, downsampling, and activation configuration, the second layer (G_0^1 and G_1^1) and the first layer (G_0^2 and G_1^2) of low-level features were calculated for I_0 and I_1 , respectively. A brief illustration of the aforementioned processes was shown below:

$$\begin{aligned} G_j^0 &= H(I_j) \\ G_j^1 &= D(G_j^0) \\ G_j^2 &= D(G_j^1) \end{aligned} \quad (\text{Equation 1})$$

where H denotes a stack of convolution and activation functions, while D denotes integration of H with a further downsampling process, with j either 0 or 1.

The Cross-Scale Feature Fusion (CSFF) module

This module functioned in fusing the vascular and non-vascular features at varied scales and promoting the representation of underlying basic features. This module was applied to I_0 and I_1 to facilitate the fusion of cross-scale features. In particular, the low-level features of the i -th layer, G_0^i and G_1^i , 2^{i-1} atrous convolutions were utilized to generate fused features⁴⁷ (with a fixed kernel size of 3, stride of 2^i , and for the n -th atrous convolution the padding and dilation were both of size n). This can be illustrated as follows:

$$F(G_j^i) = (A_1(G_j^i), \dots, A_n(G_j^i)), \quad (\text{Equation 2})$$

where F and A denotes feature fusion and atrous convolution, respectively. The number of atrous convolutions was represented by variable n . The variable n took a value of 2^{i-1} for i equivalent to 0, 1, or 2.

Furthermore, through the amalgamation of fused features from distinct scales and the adoption of linear mapping, the F_0 and F_1 cross-scale features can be derived from I_0 and I_1 , respectively. This can be expressed as:

$$F_i = \tau \left[C \left(F \left(G_i^0 \right), F \left(G_i^1 \right), F \left(G_i^2 \right) \right) \right], \quad (\text{Equation 3})$$

where τ and C denote the linear mapping and concatenation operation, respectively. Then, the fused F_i underwent flattening and normalization before being utilized as inputs for the subsequent process of the MSFE.

The Motion-Structure Feature Extractor (MSFE) module

Concatenation of fused features F_0 and F_1 produced $F_a \in R^{[n] \times d}$. To obtain this $F'_a \in R^{[n] \times d}$, a reverse concatenation was executed. This can be expressed as follows:

$$\begin{aligned} F_a &= C(F_0, F_1) \\ F'_a &= C(F_1, F_0) \end{aligned} \quad (\text{Equation 4})$$

A Lambda Layer was employed to carry out content and position-based contextual interaction simulations in a fully convolutional manner. The variables k and v were utilized to denote the depths of query and value, respectively, and $P \in R^{[n] \times d}$ were employed to denote positional information. To calculate the queries, keys, and values, the following formulae were utilized:

$$\begin{aligned} Q &= F_a W_Q \in R^{[n] \times |k|} \\ K &= F'_a W_K \in R^{[n] \times |k|} \\ V &= F'_a W_V \in R^{[n] \times |v|} \end{aligned} \quad (\text{Equation 5})$$

The $B \in R^{[n] \times |k|}$ was then utilized to denote the relative position embeddings. The normalization of the key can generate the $\bar{K} = \text{softmax}(K, \text{axis} = n)$. Next, using the following formula:

$$\begin{aligned} \lambda_c &= \bar{K}^T V \in R^{|k| \times |v|} \\ \lambda_p &= B^T V \in R^{|k| \times |v|} \end{aligned} \quad (\text{Equation 6})$$

the content-based contextual interactions $\phi_t \omega_t = \varepsilon(C(M_{0 \rightarrow t}, M_{1 \rightarrow t}, S_0, \text{ and position-based contextual interactions } \lambda_p \text{ can be calculated.})$

Ultimately, the computation of key general motion as well as structural features necessary for inferring any intermediate frame was executed by applying contextual interactions to the Q queries and P , as shown below:

$$\begin{aligned} S &= Q \lambda_c + Q \lambda_p = C(S_0, S_1) \\ M &= P \lambda_c + P \lambda_p = C(M_0, M_1) \end{aligned} \quad (\text{Equation 7})$$

Decoding and refining for multiple intermediate frames

To further obtain motion features corresponding to multiple different intermediate times t , we multiplied each t with the general motion features to map and obtain $M_{0 \rightarrow t}$ and $M_{1 \rightarrow t}$.

The Flow-Mask Estimator (FME) module

For a specific time t , $M_{0 \rightarrow t}$ and $M_{1 \rightarrow t}$ were concatenated with S_0 and S_1 , respectively, and this combination served as part a of the input for the FME, while part b , the concatenation of I_0 and I_1 , constituted the second input for the FME. Specifically, ε was used to represent FME and utilized a Pixel Shuffle upsampling and downsampling to part a and part b , respectively. Afterward, these parts were amalgamated and subjected to continuous convolution operations, resulting in the production of a two-directional optical flow ϕ_t and the mask ω_t , at a specific time t via upsampling. The aforementioned process can be formulated as:

$$\phi_t, \omega_t = \varepsilon(C(M_{0 \rightarrow t}, M_{1 \rightarrow t}, S_0, S_1), C(I_0, I_1)). \quad (\text{Equation 8})$$

Thereafter, ϕ_t was adopted to warp I_0, I_1 , the low-level features G_i^j from various layers derived with the aid of the FE module, and the general structural features S_0 and S_1 derived via the MFSE module. The findings following were expressed in the following format. For instance, X_2^y was expressed as $\tilde{X}_2^y$ following warping. Subsequently, $I_0, I_1, \tilde{I}_0, \tilde{I}_1$ were concatenated.

The Refiner module

Finally, features at different scales were incorporated and refined into O_t in a layer-wise manner by a simplified UNet⁴⁸ during the process of the Refiner. Then, this was accompanied by a skip connection. The intermediate frame $\tilde{I}_t$ that matches with t was afterward derived using:

$$\hat{I}_t = \tilde{I}_t + R(O_t, G, S) \quad (\text{Equation 9})$$

With R , G , and S representing the Refiner, the collection of G_i^j , and the collection of S_0 and S_1 , respectively. To determine $\tilde{I}_t$, the following expression was used:

$$\tilde{I}_t = \omega_t \odot \text{backwarp}(I_0, \phi_{t \rightarrow 0}) + (1 - \omega_t) \odot \text{backwarp}(I_1, \phi_{t \rightarrow 1}). \quad (\text{Equation 10})$$

where the symbol $\odot$ denotes the Hadamard product.

Loss functions

An integration of three types of loss functions was implemented, as shown below:

$$L = \omega_1 L_1 + \omega_{VGG} L_{VGG} + \omega_{Style} L_{Style}, \quad (\text{Equation 11})$$

where L_1 represents the L1 reconstruction loss. To promote texture quality and improve image details, L_{VGG} integrated the L1 norm of the VGG-19 features. Further, the style loss L_{Style} employed the L2 norm of the automatic correlation of the VGG-19 features^{49,50}. In summary, this approach intended to ascertain the benefits of L_{VGG} in replicating and capturing style patterns and textures in a more effective manner. Pertaining to the selection of the weights (ω_1 , ω_{VGG} , ω_{Style}), it can be referred to the ref. 33.

Dataset preparation and preprocessing

A total of 29988 videos (sequences) of three datasets (datasets 1-3) were utilized for the current study, and the baseline information of those 3 datasets was shown in Table 1. Those sequences were comprised of a variety of image resolutions, including 489×489 , 500×500 , 512×512 , 480×616 , 952×952 , 742×960 , 960×960 , 960×1024 , and 1024×1024 ; they covered various parts of human body, including head, thorax, abdomen, limb, and pelvis, which were collected by using 2D- and 3D- DSA. We specifically categorized the tasks into 8 major categories based on imaging modality and body part: 2D all parts, 2D head, 2D thorax, 2D abdomen, 2D limb, 2D pelvis, 3D vascular category (including 3D head, 3D pelvis), 3D non-vascular category (including 3D thorax, 3D abdomen, 3D limb, 3D pelvis). The 1st major category (i.e., 2D all parts) was used for pre-training and the remaining seven categories were used for fine-tuning and testing. The above tasks were further divided into $8 \times 3 = 24$ subcategories based on the 3 different number of frame generations (i.e., 1 to 3 frames).

Model configuration

For generating 1 to 3 frames, time t sequences were set to [0.5], [0.33, 0.67], and [0.25, 0.50, 0.75], respectively. For simulating contextual interactions, context modeling scope r sizes were 29, 29, and 21, respectively. Effects of varying r on the final results were compared in the ablation study.

Pretraining details

For the 8 major classes of generation tasks, we initially aggregated and pre-trained on 50% of the data from all body parts in 2D from 7 hospitals (i.e., centers 1–7). The obtained weights were then fine-tuned to complete the tasks of the other 7 major classes. Specifically, those models were trained on 8 NVIDIA A100 GPUs (40 GB), and the batch sizes were set to 10, 10, and 6 for tasks interpolating 1 to 3 frames, with warm-up steps set to 26972, 28486, and 40348, respectively. Meanwhile, The AdamW⁵¹ optimizer was employed with $\beta_1 = 0.9$, $\beta_2 = 0.999$, as well as a weight decay of $1e - 4$. In addition, the learning rate was warmed up to $2e - 4$ and then decayed following a cosine schedule, decreasing to $2e - 5$ over 300 epochs. Each frame was cropped to a fixed resolution of 320×320 and random flip and rotation were implemented for augmentation. Regarding the selection of loss weights (ω_1 , ω_{VGG} , ω_{Style}), it is referred to the ref. 33; specifically the weights of (1.0, 1.0, 0.0) were assigned to the first epoch, whereas the weights of (1.0, 0.25, 40.0) were applied to the subsequent epochs.

Fine-tuning details

Subsequently, the weights obtained from the above pretraining were fine-tuned. Specifically, for the 5 major classes of tasks involving 2D body parts (head, thorax, abdomen, limb, and pelvis), the remaining 50% data from various parts of the respective hospitals (centers 1–7) were utilized. Since 3D scanning is a non-essential operation in the vast majority of clinical cases, much less 3D data were obtained as compared to 2D data, which is insufficient for fine-tuning each body part, so we only categorized two 3D fine-tuned models into vascular and non-vascular. For the 2 major classes related to 3D vascular and non-vascular tasks, all data from centers 1–7 were employed. Most technical details remained consistent with the pretraining phase, with the following 2 exceptions: (1) A smaller learning rate was utilized to search for optimal weights during the period of fine-tuning. The learning rate was warmed up to $2e - 5$ and then decayed using cosine annealing, reaching $2e - 6$ over 300 epochs. (2) Considering the varying data volumes and batch sizes among different tasks, we meticulously adjusted the warm-up steps for effective model preheating, as shown in Table S7. The convergence times were as follows: for 2D fine-tuning tasks, within 13 h; for 3D non-vascular fine-tuning tasks, within 133 h; and for 3D vascular fine-tuning tasks, within 16 h.

Testing details

The seven fine-tuned models (2D: head, thorax, abdomen, limb, and pelvis; 3D: vascular, non-vascular) were tested with data from three other hospitals (centers 8–10). All tests were executed on a single NVIDIA A100 GPU (40 GB).

Model comparisons

To highlight the advantages of the proposed method in DSA frame generation, we conducted quantitative and visual comparisons with other advanced video frame generation methods (EMA-VFI,³² FILM³³) on the task of generating 2 frames. We faithfully adhered to the fully sampled configurations, retrained, and fine-tuned them on our dataset. For the proposed method, two specific versions

were trained: MoStNet- $\mathcal{L}_1$, employing only $\mathcal{L}_1$ loss to achieve higher test scores, whereas MoStNet, utilized the proposed combined loss $\mathcal{L}$ for improved image quality. When evaluating visual effects, the version of the model that yielded highest image quality were employed, i.e., the proposed MoStNet, and FILM- $\mathcal{L}_{style}$.

Inclusion and exclusion criteria

Inclusion Criteria.

- (1) Underwent DSA examination, and obtained a series of DSA examination video sequences;
- (2) Scanning equipment used was a worldwide used DSA machine (Siemens, Philips Healthcare, GE HealthCare).

Exclusion Criteria.

- (1) Number of images in the DSA examination sequence was ≤ 10 ;
- (2) Poor image quality in DSA images, preventing observation and assessment.

To enhance the diversity of samples, we selected the most representative sequence from multiple examination sequences for each patient.

Subjective evaluations

Prior to video analysis, the readers underwent a training session to familiarize themselves with the Likert-scale classification. Each reader was given a dataset with different grading scores for each analyzed videos quality-based outcome. Prior to conducting the experiments, the details of the patients in the videos were hidden. The whole process of subjective evaluation of the generated videos was divided into three aspects.

Visual Turing Tests

To corroborate the performance of the proposed model, a type of VTTs (Visual Turing Tests) was designed, which was performed by 5 board-certified interventional radiologists. In the VTTs, participating interventional radiologists were instructed to differentiate the generated videos from the frame full-sampled ones. Based on the five body parts (due to the lack of 3D data for the pelvis and limbs, these two datasets were combined into one set), along with two undersampling rates and two types of dimensions of videos, a total of 18 sets of VTTs were designed to evaluate the performance of generation models with different capabilities, with each set consisting of 50% fully sampled videos and 50% generated videos.

Overall video quality

Five interventional radiologists (R1 to R5) independently examined the generated videos. The generated videos with 1- and 2-frame generations were used to evaluate the video quality. The evaluation criteria were presented in the [STAR Methods](#). The video scores obtained from different body parts of the fully-sampled videos and generated videos with 1- and 2-frame generations (both 2D and 3D datasets) were used to assess the reproducibility of the scoring method.

Diagnostic confidence for lesions

The generated videos with 1- and 2-frame generations were used for evaluating the diagnostic confidence, with the clinical reports served as the gold standard to determine whether lesions present. The following scoring standards were utilized to determine the diagnostic confidence for lesions: 1 point = very uncertain; 2 points = not so sure; 3 points = neutral; 4 points = basically certain; 5 points = very sure. Notably, scores ≥ 4 points were considered as high diagnostic confidence.

Finally, we conducted a consistency assessment for the 18 sets of Visual Turing Tests, 49 sets of overall video quality evaluations, and 49 sets of lesions diagnostic confidence assessments.

Likert-scale scoring criteria for evaluating DSA video quality

1 point: The video exhibits very poor smoothness, with the animation significantly stuttering, hardly providing a seamless viewing experience.

2 points: The video exhibits poor smoothness, with the animation relatively lagging, providing an overall subpar viewing experience.

3 points: The video exhibits moderate smoothness, with the animation relatively smooth, providing a satisfactory level of viewing experience.

4 points: The video exhibits a high level of smoothness, with the animation smooth and natural, providing an excellent overall viewing experience.

5 points: The video exhibits an extremely high level of smoothness, with the animation exceptionally fluid, providing an outstanding viewing experience.

Objective evaluations

Two key quantitative indicators (SSIM and PSNR) were used to assess the similarities and differences between our results and the gold standard. Specifically, SSIM score was used for measuring the overall similarity between the two images, and PSNR was defined as the ratio between the maximum signal power and the noise power that affects the image quality. In general, a higher

SSIM and PSNR score indicates a better generation result, which can be expressed as:

$$\text{PSNR} = 20 \log_{10} \frac{\max(\text{ori}) \sqrt{N}}{\|\text{ori} - \text{int}\|_2^2} \quad (\text{Equation 12})$$

$$\text{SSIM} = \frac{2\mu_{\text{ori}}\mu_{\text{int}} + c_1}{\mu_{\text{ori}}^2 + \mu_{\text{int}}^2 + c_1} \cdot \frac{2\sigma_{\text{ori_int}} + c_2}{\sigma_{\text{ori}}^2 + \sigma_{\text{int}}^2 + c_2} \cdot \frac{\sigma_{\text{ori_int}} + c_3}{\sigma_{\text{ori}}\sigma_{\text{int}} + c_3} \quad (\text{Equation 13})$$

with *int* representing the generated frame, *ori* denoting the fully sampled frame, and *N* being the total number of image voxels. The SSIM index was a multiplicative combination of the luminance term, the contrast term, and the structural term. The values of μ_{int} and μ_{ori} were the mean values of generated and fully sampled frames, respectively. The values of σ_{int} and σ_{ori} were the standard deviations of generated and fully sampled frames, respectively. The value of $\sigma_{\text{int_ori}}$ denoted the covariance of the generated and the fully sampled frames. The values of c_1 , c_2 , and c_3 were the non-negative real numbers that specified the regularization constants for the luminance, contrast, and structural terms, respectively.

QUANTIFICATION AND STATISTICAL ANALYSIS

The performance of the proposed models was assessed using both subjective and objective evaluation indicators. According to the Likert scale classification, we scored the performance of the proposed models with respect to the fully sampled videos and generated videos with 1- and 2- frame generations for both 2D- and 3D-DSA.

Subjective evaluations

Kendall coefficient of concordance (*W*) was used to assess the scoring consistency of the 5 readers (the scoring method used for evaluating the fully sampled and generated videos and the lesions diagnostic confidence were also studied). The strength of agreement between the observers was classified as poor ($W < 0.2$), fair ($0.2 \leq W < 0.4$), moderate ($0.4 \leq W < 0.6$), strong ($0.6 \leq W < 0.8$), or super ($0.8 \leq W < 1.0$). Fleiss's Kappa was used to assess the inter-reader reliability among multiple interventional radiologists for assessing authenticity of the generated videos obtained with the proposed model. The strength of agreement for κ values is as follows: less than 0.20, poor; 0.21–0.40, fair; 0.41–0.60, moderate; 0.61–0.80, substantial; or 0.81–1.00, almost perfect agreement. ANOVA tests were conducted to compare the video quality scores obtained using the Likert scale classification method (scores were derived from the average score of the 5 readers). The Bonferroni method was used to control for the probability of false positives, and the results were compared. Notably, all the scores of image quality and diagnostic confidence were presented as mean $\pm$ SD. Pairwise comparisons between any videos (fully sampled and generated videos with 1- and 2-frame generations) were performed following a post hoc analysis method, followed by Bonferroni correction. ANOVA tests were conducted to compare the subjective indices and the diagnostic confidence, whereas the Kendall coefficient of concordance (*W*) tests were conducted to determine the consistency of the diagnostic confidence obtained by different observers. In the current study, all $p < 0.05$ were considered statistically significant.

Objective evaluations

PSNR and SSIM values of each frame were calculated and mean $\pm$ SD were tallied according to the different groups.